In [7]:
```
import numpy as np
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import confusion_matrix, accuracy_score, precision_score, r
```

In [14]:
```
data = pd.read_csv('Iris.csv')
```

In [15]:
```
X = data.iloc[:, :-1]
y = data.iloc[:, -1]
```

In [16]:
```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42)
```

In [17]:
```
# Train model
model = GaussianNB()
model.fit(X_train, y_train)
```

Out[17]:
```
▾ GaussianNB
▶ Parameters
```

In [18]:
```
y_pred = model.predict(X_test)
```

In [19]:
```
cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:\n", cm)
```

```
Confusion Matrix:
 [[19  0  0]
 [ 0 12  1]
 [ 0  0 13]]
```

In [20]:
```
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred, average='macro')
recall = recall_score(y_test, y_pred, average='macro')
f1 = f1_score(y_test, y_pred, average='macro')

print("Accuracy:", accuracy)
print("Precision:", precision)
print("Recall:", recall)
print("F1 Score:", f1)
```

```
Accuracy: 0.9777777777777777
Precision: 0.9761904761904763
Recall: 0.9743589743589745
F1 Score: 0.974320987654321
```

In [ ]: